Angel Ivan Rodriguez Leon (Phd student)
Chemistry Department, Northwestern University
angelrodriguezleon2029@u.northwestern.edu

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Argonne Training Program on Extreme-Scale Computing (ATPESC) 2026

The Argonne Training Program on Extreme-Scale Computing (ATPESC) provides an intensive, two-week training on the key skills, approaches and tools to design, implement and execute computational science and engineering applications on current high-end computing systems and the leadership-class computing systems of the future.. This document should include your purpose for attending the training program, a description of your computing experience, and your current/future CSE research or RSE plans. This part of the application should be no longer than two pages. Please include your first name, last name, job title, and institution at the top of the document. NOTE: The training program's curriculum is geared for participants with substantial MPI, OpenMP, and/or Data Science framework programming experience who have used an HPC system for a reasonably complex application and have conducted or are preparing to conduct CSE research on large-scale computers. Please describe in detail your previous experience with HPC systems (e.g., size of simulations, kind of systems, experience as user and/or developer, software and solvers used, etc.) and your future CSE or RSE plans on leadership-class supercomputers. Specific details about your experience in relation to the eligibility criteria are essential to a successful application. Be clear about your personal contributions to any group projects. Additional topics to be covered include computer architectures, mathematical models and numerical algorithms, approaches to building community codes for HPC systems, and methodologies and tools relevant for Big Data applications. Computer architectures and predicted evolution Numerical algorithms and mathematical software Software productivity and sustainability Data analysis, visualization, I/O, and methodologies and tools for big data applications Performance measurement and debugging tools Machine learning and data science. Substantial experience in MPI, OpenMP, and/or Data Science Frameworks- MAKE EMPHASIS ON FORMAL TRAINING, AND EXTREME PART. BEING EABLE TO LEARN FROM LEADERS AND FROM THE RESOURCES OF NATIONAL LAB

Other

I'm doing my Ph.D. in Chemistry at Northwestern University, where I focus on computational simulations. I've got a background in electronic structure calculations and molecular dynamics simulations. I work with both Prof. Monica Olvera de la Cruz at Northwestern and Prof. Maria K. Chan at Argonne National Lab. I'm especially excited about AI for Science and want to focus on using machine learning to transform how we discover, understand, and design new materials and molecules.
As a mathematical physicist student, I have learned and understood the impacts of scientific and technological innovations on society's development. Therefore, my professional work focuses on the implementation of several scientific fields to solve current issues with strong social consequences. In particular, I want to research molecular systems at the atomistic level, with the tools of computational physics and chemistry, within the framework of first principle theories. Despite the constant improvement of these theories, like density functional theory and several variational and perturbative methods, they still have several deficiencies. The high computational cost and size restrictions of the arrangements demand innovative solutions. I want to learn how to implement these models with the aid of novel statistical theories, artificial intelligence, and machine learning toward computational efficiency. I believe that the Argonne Training Program on Extreme-Scale Computing could help me achieve the knowledge and experiences necessary to enrich my professional objectives From my experience, computational technologies are indispensable for natural science research. Rapid improvement of these technologies demands us to keep pace with cutting-edge methods and learn how to implement them in meaningful investigations, which is by no means trivial. During my academic formation, I performed theoretical studies that fall into an area where different scientific fields coexist, like biology, chemistry, physics, and computer science. In recent times, there has been increasing attention to molecules employed in the construction of organic solar cells to solve global warming problems.
Particularly, for my bachelor's degree thesis, I performed a theoretical study on polymers directly involved in

sustainability issues by characterizing their electronic, optical, and magnetic properties. The analysis was under the scope of DFT using ADF software. During my undergraduate studies at Instituto Politecnico Nacional, I conducted a theoretical study of molecules used for organic solar cells, guided by Dr. Juan Ignacio Rodriguez Hernandez. The study, titled "Theoretical study of polymers widely used in fullerene free organic solar cells: a quantum chemistry computational analysis" focused on the polymer PBDB-T-SF and small molecule IT-4F. Using the Amsterdam Density Functional (ADF) software and by accessing one of the University of Buffalo supercomputers, I performed density functional theory including geometry optimizations and electronic assessments based on HOMO and LUMO values, as well as energy gaps. Additionally, I conducted TD-DFT calculations to characterize their optical properties, examining electron transitions and UV/vis spectra. The study revealed highly localized fragments within the molecules for the first five HOMO and LUMO levels and indications of fragment-to-fragment transitions. These findings suggested the potential of fullerene-free organic solar cells to enhance solar energy efficiency. The project not only demanded an understanding of solar energy devices but also required me to learn how to remotely access supercomputers for performing complex calculations. This work allowed me to contribute to sustainable energy advancements while engaging in a global concern that demands international cooperation.

Additionally, in biological physics, the behavior of DNA during diverse cellular processes and its interaction with other molecules is of paramount importance to elucidate genetic diseases, new nanotechnology techniques, and drug discovery, among others. Previous works had failed to include the quantum mechanical effects of these systems, like bond breakage and charge transfer. My research during my master's degree focused on first principle molecular dynamics simulations on short ds-DNA molecules. The study aimed to analyze the mechanical properties of DNA under harmonic external forces. The structures are immersed in an implicit Langevin solvent to represent realistic cellular conditions. Helix dissociation was achieved with the use of virtual springs attached to specific sites and implemented in a computer code. The springs move sequentially along the nucleic strands, simulating the enzymatic behavior of helicases. Additionally, during my master's at UNAM, I joined Dr. Ruben Santamaria's molecular simulations group. I wanted to investigate biomolecular systems. Dr. Santamaria proposed to analyze DNA behavior under forces that approximate replication processes using molecular dynamics. Despite my background in mathematical physics and quantum chemistry, I lacked biological expertise for this study. Moreover, the computational cost of the simulations required supercomputers and advanced programming. To bridge my knowledge gap, I attended biological classes and enrolled in a high-performance computing certificate program. This experience gave me the necessary tools to tackle this project. Previous studies on DNA under external forces simplified the DNA structures for cost-efficiency or used classical simulations, which failed to include crucial aspects. Recognizing the significance of studying the complete DNA structure, I combined a quantum mechanical treatment with a classical approach to evolve the molecular system in time. The electronic calculations were performed by using the TeraChem software and the nuclei dynamics with a computational code from our group. I developed a code to smoothly open the double-helix, approximating the helicase enzymatic action with harmonic forces. Each base pair is opened until a specific distance when we can safely assume the hydrogen bonds have been broken and move sequentially to the next base pair along the nucleic strand. Considering the cellular environment, we employed an implicit Langevin aqueous model, accounting for thermal agitation and solvent viscosity. I had the opportunity to access the MIZTLI supercomputer at UNAM to perform the calculations. This supercomputer is one of the fastest in Mexico, and even with those resources, opening an entire five-pair-long DNA molecule took about a month of calculation time. This research calculates nucleic acid base separation forces, considering the exact position and stacking interactions of each base pair. The level of theory allows for analysis of electric charge behavior during the opening of the double helix. The model takes into account the complete phosphate backbone and complements experimental DNA studies. I presented this work at the "National Physics Congress" in Mexico and the "Gulf Coast Undergraduate Research Symposium" at Rice University. This study was recently sent for publication. Understanding these fundamental aspects of biomolecules during cellular processes holds great promise in shedding light on genetic diseases, drug design, and advancing nanotechnology techniques.

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Therefore, I am looking for the right academic program that can teach me new abilities in the field of theoretical and computational sciences to analyze molecular systems and help me to broaden my perspective as a young researcher in an eclectic and ever-changing world. In particular, professor Bingqing Cheng's group has been doing some exciting and inspiring research in this direction. I believe that the interests and vision of the group are a good match for my academic background and my future career goals and would be an honor to work with such outstanding scientists. I consider myself a sedulous student with the desire to keep learning and expanding my experiences. I intend to apply for a Ph.D. position in a German and Austrian university because of my desire to become a formal researcher and be able to do it with the motivation and inspiration of a beautiful culture. I think this internship would be an excellent first encounter toward this goal. I am excited to be considered and I am hoping to have the opportunity to continue enjoying, growing, and learning with your help.
For instance, the projects required me to access supercomputers remotely from both Mexico and the United States. This experience made me realize my limited understanding of the whole process, driving me to enroll in a high-performance computing certificate program, to better understand the relation between scientific problems and the construction of computational solutions. I intend to improve my knowledge even further by pursuing my graduate studies in the US.
Despite these experiences, there remain crucial aspects of molecular-level behavior in medicine and biology that I aspire to comprehend more deeply, which hold great promise in shedding light on genetic diseases, drug design, and advancing nanotechnology techniques. For example, how protein misfolding contributes to diseases, how protein-ligand interactions can inhibit or enhance the function of the protein, or how self-assembly molecules can act as nanovehicles. I want to employ diverse fundamental physical and chemistry models, coupled with advanced programming, like AI and Machine Learning to study these types of phenomena. These novel techniques have the potential to accelerate calculations by predicting potential energy surfaces and electronic structures, predicting solvent behavior, and analyzing more complex systems with less computational cost. The computational resources and collaborations with many research centers offer a great opportunity to expand my skills in theoretical and computational sciences and help me broaden my perspective as a young researcher in an eclectic and ever-changing world.

I consider myself a dedicated person with the desire to keep learning and my previous work and personal background make me a qualified Ph.D. candidate. I dream of establishing a molecular simulation lab and applying the acquired knowledge to interesting problems. I am excited to be considered and I am hoping to have the opportunity to continue enjoying, growing, and learning with the guidance and support of your institution. Let's continue to be curious and stubborn together.